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Project:Part 1

A. Main Entities

Outlet

Represents the different rental locations of the different companies where staff are assigned and vehicles are stored.

Vehicle

Shows the vehicle available for hiring and each one is identified by registration number.

Client

Customers who hire vehicles from companies.

Hire_Agreement

Shows the transactions between client and company.

Staff

Represents employees working at different outlets of the company.

B. Main Relationship between Entities

Main relationship between entities are

Outlet-Vehicle

Outlet stores Vehicles.

Outlet-Staff

Outlet hires or employs staff.

Client-Hire_Agreement

Client enters the hire agreement.

Vehicle-Hire_Agreement

Vehicles are assigned to the hire agreements.

C. Multiplicity Constraints

Outlet : Vehicle

One Outlet has many vehicles while each Vehicle belongs to an Outlet.

Outlet : Staff

One Outlet employs more than one Staff member while every Staff member works at one Outlet.

Client : Hire_Agreement

One Client can have more than one Hire_Agreements while each Hire_Agreement belongs to one Client.

Vehicle : Hire_Agreement

One Vehicle can be assigned to a Hire_Agreement more than once while each Hire_Agreement covers one Vehicle.

D. Attributes

Outlet

Outlet_number
Address
Phone_number
Fax_number

Vehicle

Registration_number
Make
Model
Engine_size
Capacity
Current_mileage
Daily_hire_rate
Current_location

Staff

Staff_number
First_name
Last_name
Home_address
DOB
Sex
Date_joined
Job_title
Salary

Hire_agreement

Hire_number
Date_started
Date_wish_end
Mileage_before
Mileage_after

Client

Client_number
First_name
Last_name
Address
Phone_number
DOB
Driver_license_number

E. Candidate and Primary Key**Outlet**

Candidate Key: Outlet_number
Primary Key: Outlet_number

Vehicle

Candidate Key: Registration_number
Primary Key: Registration_number

Staff

Candidate Key: Staff_number

Primary Key: Staff_number

Hire_agreement

Candidate Key: Hire_number

Primary Key: Hire_number

Client

Candidate Key: Client_number, Driver_license_number

Primary Key: Client_number

F. ER-diagram

